UNIVERSITÉ Concordia UNIVERSITY CONTINUING EDUCATION

MOD10: Introduction to Machine Learning

Project — Deliverable 2

Data exploration & preprocessing

Due: Lesson 3 (before class). **Groups:** same 2–3 as Deliverable 1. **Submit:** one notebook (.ipynb) per group on Moodle.

Goal

Take the dataset you chose in Deliverable 1 and make it **model-ready**. You'll explore it, fix what's broken, and shape it so a machine-learning model can use it next week. This is the first **coding** deliverable: you submit a Jupyter notebook, not a written summary.

Note: the Class 2 lab worksheet is practice on a sample dataset and is not graded. Deliverable 2 is the graded work — the same steps applied to *your own* dataset.

What your notebook must contain

Organize it with a short markdown note before each step explaining what you did and why.

1. **Exploration (EDA)** — `.head()`, `.shape`, `.info()`, `.describe()`, at least **one plot**, and **2–3 observations** about the data.
2. **Missing values** — report how many (`.isna().sum()`), then **fill or drop** each, with a one-line reason.
3. **Encoding** — make every feature numeric: `map()` for binary/ordered columns, `get_dummies()` for unordered ones.
4. **Scaling** — standardize the numeric features (e.g. `StandardScaler`).
5. **Train/test split** — separate `X` and `y`, then `train_test_split` (use `stratify=y` for classification; fit any scaler on `X_train` only).
6. **Short reflection** (a markdown cell) — the main decisions you made and anything that surprised you.

Submission checklist

- ☐ **Group members** listed at the top of the notebook.
- ☐ The notebook **runs top to bottom without errors** (`Kernel → Restart & Run All`).
- ☐ **All five steps present**, each with a short markdown explanation.
- ☐ **No missing values and no text feature columns remain** after preprocessing.
- ☐ One `.ipynb` submitted per group on Moodle (plus the dataset, or a link to it).

How it will be assessed

Graded as part of the project's weekly deliverables. We look for:

- Preprocessing **done correctly** — missing values handled, features numeric, data scaled and split.
- Choices are **justified**, not just applied — you can say why median vs. drop, map vs. one-hot.
- A **clean, runnable notebook** with clear markdown notes.
- Evidence you **understood your data** — meaningful EDA observations.

Tips

- Reuse the Class 2 lab as a template — the steps are identical, only the dataset changes.
- Fit the scaler on the **training set only** — this avoids data leakage (we'll return to this).
- Keep an ID column (if any) out of the features — it isn't something to learn from.

Stuck on your dataset? Bring it to class or email me — preprocessing is where most real-world ML time goes, so questions are expected.